# MA2031- Linear Algebra for Engineers

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Module-5 (Lecture note-2)

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# Recall: diagonalizability

Let $\mathcal{V}$ be a finite dimensional $\mathbb{F}$-vector space and $T \in \mathcal{L}(\mathcal{V})$. Given a basis $\mathcal{B}$ for $\mathcal{V}$ we can always define $A := {}_{\mathcal{B}}[T]_{\mathcal{B}} \in M_n(\mathbb{F})$.

**Question:** Can we get a basis $\widetilde{\mathcal{B}}$ such that ${}_{\widetilde{\mathcal{B}}}[T]_{\widetilde{\mathcal{B}}}$ is diagonal?

If yes, then

$$ {}_{\mathcal{B}}[T]_{\mathcal{B}} = Q \; {}_{\widetilde{\mathcal{B}}}[T]_{\widetilde{\mathcal{B}}} \; Q^{-1}. $$

for some invertible $Q \in M_n(\mathbb{F})$. So

$$ {}_{\widetilde{\mathcal{B}}}[T]_{\widetilde{\mathcal{B}}} \text{ diagonal } \longleftrightarrow A = QDQ^{-1} \text{ for some diagonal matrix } D \in M_n(\mathbb{F}) $$

**Question:** Given $A \in M_n(\mathbb{F})$ when it is diagonalizable?

Every $A \in M_n(\mathbb{F})$ is not always diagonalizable. We proved necessary and sufficient conditions for diagonalizability.

## Unitary diagonalization

Let $\mathcal{V}$ be a finite dimensional **IPS** and $T \in \mathcal{L}(\mathcal{V})$.

**Question:** Can we get an orthonormal basis $\mathcal{B}$ such that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is diagonal?

- Recall: Let $Q \in M_n(\mathbb{F})$. Then $Q$ is invertible iff its columns are linearly independent (and hence a basis for $\mathbb{F}^n$.)
- Exercise: Suppose $U \in M_n(\mathbb{F})$. Show that $U$ is a unitary iff the columns of $U$ are orthonormal vectors (and hence a orthonormal basis for $\mathbb{F}^n$.)

So the above question is equivalent to the following:

**Question:** Given $A \in M_n(\mathbb{F})$ when does there exists a unitary $U \in M_n(\mathbb{F})$ and a diagonal $D \in M_n(\mathbb{F})$ such that $A = UDU^*$?

**Definition:** $A \in M_n(\mathbb{F})$ is said to be unitarily diagonalizable if there exists a unitary $U \in M_n(\mathbb{F})$ such that $U^*AU$ is a diagonal matrix.

# Schur's upper triangularization

**Theorem: (Schur)**

1. Suppose $\mathcal{V}$ is a finite dimensional **complex IPS** and $T \in L(\mathcal{V})$. Then there exists an orthonormal basis $\mathcal{B}$ such that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is an upper-triangular matrix.
2. Given $A \in M_n(\mathbb{C})$ there always exist a unitary $U \in M_n(\mathbb{C})$ such that $U^*AU$ is an upper triangular matrix.

**Proof of (2):**

- Proof by induction.
- Clearly true for $n{=}1$. Assume the theorem up to $n-1$ with $n \geq 2$.
- Suppose $d_1$ is an eigen value of $A$ and $v_1$ is a corresponding eigen vector.
- We assume $\|v_1\| = 1$. Otherwise replace $v_1$ by $\frac{v_1}{\|v_1\|}$.
- Extend $v_1$ to an ONB $\{v_1, v_2, \cdots, v_n\}$ for $\mathbb{C}^n$.
- Then $V = \left[v_1|v_2|\cdots|v_n\right] \in M_n(\mathbb{C})$ is unitary.

## Proof continues...

- $AV = \left[Av_1|Av_2|\cdots|Av_n\right] = \left[d_1v_1|Av_2|\cdots|Av_n\right]$
  $= V\left[\ x|V^*Av_2|\cdots|V^*Av_n\right]$ where $x = (d_1, 0, \cdots, 0)^t \in \mathbb{C}^n$.
- Thus $A = V\begin{bmatrix} d_1 & R \\ 0 & \widetilde{A} \end{bmatrix}V^*$ where $0 \in \mathbb{C}^{n-1}, R \in M_{1\times(n-1)}(\mathbb{C}), \widetilde{A} \in M_{n-1}(\mathbb{C})$.
- $|A - \lambda I| = |V(\begin{bmatrix} d_1 & R \\ 0 & \widetilde{A} \end{bmatrix} - \lambda I)V^*| = |\begin{bmatrix} d_1 - \lambda & R \\ 0 & \widetilde{A} - \lambda I \end{bmatrix}| = |d_1 - \lambda||\widetilde{A} - \lambda I|$
  i.e., $\sigma(A) = \sigma(\widetilde{A}) \cup \{d_1\}$.
- By assumption there exists unitary $\widetilde{V}$ and upper triangular matrix $\widetilde{B}$ in $M_{n-1}(\mathbb{C})$ such that $\widetilde{A} = \widetilde{V}\widetilde{B}\widetilde{V}^*$.
- Note that $V' = \begin{bmatrix} 1 & 0 \\ 0 & \widetilde{V} \end{bmatrix}$ and $U = VV'$ are unitaries in $M_n(\mathbb{C})$.
- $U^*AU = U^*V\begin{bmatrix} d_1 & R \\ 0 & \widetilde{A} \end{bmatrix}V^*U = V'^*V^*V\begin{bmatrix} d_1 & R \\ 0 & \widetilde{A} \end{bmatrix}V^*VV'$
  $= V'^*\begin{bmatrix} d_1 & R \\ 0 & \widetilde{A} \end{bmatrix}V' = \begin{bmatrix} 1 & 0 \\ 0 & \widetilde{V}^* \end{bmatrix}\begin{bmatrix} d_1 & R \\ 0 & \widetilde{A} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & \widetilde{V} \end{bmatrix}$
  $= \begin{bmatrix} 1 & * \\ 0 & \widetilde{V}^*\widetilde{A}\widetilde{V} \end{bmatrix}$
  $= \begin{bmatrix} 1 & * \\ 0 & \widetilde{B} \end{bmatrix}$ upper triangular. $\square$

**Observation:** Suppose $A$ is unitarily diagonalizable, say $A = UDU^*$. Then
$$A^*A = (UD^*U^*)(UDU^*) = UD^*DU^* = UDD^*U^* = AA^*.$$
Thus, $A$ unitarily diagonalizable implies $A$ normal. What about the converse?

**Theorem (1A):** Given a matrix $A \in M_n(\mathbb{C})$ the following conditions are equivalent:

1. $A$ is unitarily diagonalizable.
2. $A$ is normal.

**Proof:** $(1) \Rightarrow (2)$ Done !!
$(2) \Rightarrow (1)$ By Schur's theorem there exists a unitary matrix $U \in M_n(\mathbb{C})$ such that $D := U^*AU$ is upper triangular. Note that
$$DD^* = (U^*AU)(U^*A^*U) = U^*AA^*U = U^*A^*AU = D^*D.$$
Thus, $D$ is upper triangular normal matrix, hence $D$ is diagonal (verify!), i.e., $U^*AU$ is diagonal. $\square$

**Theorem (1B):** Suppose $\mathcal{V}$ is a finite dimensional **complex IPS** and $T \in L(\mathcal{V})$.

Then the following conditions are equivalent:

1. There exists an orthonomal basis $\mathcal{B}$ such that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is diagonal.
2. $T$ is normal.

## Exercises

1. Suppose $A \in M_n(\mathbb{C})$ with $\sigma(A) = \{\lambda_1, \lambda_2, \cdots, \lambda_n\}$. Show that $trace(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$ and $det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$.

   (Hint: Use Schur's theorem, and the fact that eigenvalues of an upper-triangular matrix are precisely the diagonal entries.)

2. Suppose $A \in M_n(\mathbb{C})$ is an upper triangular matrix. Show that $A$ is normal iff $A$ is a diagonal matrix.

3. Determine which of the following matrices are unitarily diagonalizable?

   (a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & -1 \\ 2 & i & 0 \\ 3 & 5 & 1 \end{bmatrix}$

4. Let $T : \mathbb{C}^2 \to \mathbb{C}^2$ be given by $T((x_1, x_2)^t) = (x_1 + x_2, x_1 - x_2)^t$. Does there exists a orthonormal basis $\mathcal{B} \subseteq \mathbb{C}^2$ such that ${}_{\mathcal{B}}[T]_{\mathcal{B}}$ is a **lower**-triangular matrix?

   (Hint: Given $\mathcal{B} = \{u_1, u_2, \cdots, u_n\}$ let $\widetilde{B} = \{u_n, \cdots, u_2, u_1\}$ and consider ${}_{\widetilde{\mathcal{B}}}[T]_{\widetilde{\mathcal{B}}}$.)